

Last login: Mon Jun 15 19:33:08 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

```
tpch=# RESET ROLE;
RESET
tpch=# SET ROLE alice;
SET
tpch=> EXPLAIN (
    ANALYZE true,
    VERBOSE true,
    COSTS true,
    BUFFERS false,
    TIMING false,
    FORMAT JSON
)
select
    o_orderpriority,
    count(*) as order_count
from
    orders
where
    o_orderdate >= date '1994-01-01'
    and o_orderdate < date '1994-01-01' + interval '3' month
    and exists (
        select
            *
        from
            lineitem
        where
            l_orderkey = o_orderkey
            and l_commitdate < l_receiptdate
    )
group by
    o_orderpriority
order by
    o_orderpriority;
WARNING: ===== planner input query tree =====
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtable 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
```

```

:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
)
:location -1
}
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {FUNCEXP
    :funcid 316
    :funcresulttype 701
    :funcset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
}
:rtekind 9
:groupexprs (
  {VAR
    :varno 1
    :varattno 6

```

```

:varstype 25
:varystmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 116
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXPRESS
:boolop and
:args (
{OPEXPRESS
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varystmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 187
}
)
{CONST
:consttype 1082
:constystmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 207
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 199
}
{OPEXPRESS
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varystmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 228
}
)
{OPEXPRESS
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:constystmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 247
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
{CONST
:consttype 1186
:constystmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 271
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 260
}
)
:location 240
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false

```

```

:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
      }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OEXPR
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 387
        }
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 400
        }
      )
      :location 398
    }
    {OEXPR
      :opno 1095
      :opfuncid 1087
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 12
          :vartype 1082
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 12
          :location 427
        }
        {VAR
          :varno 1
          :varattno 13
          :vartype 1082
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 13
          :location 442
        }
      )
      :location 440
    }
  )
  :location 423
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 325
      }
    }
  :resno 1

```

```

:resname 1_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 325
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 325
}
:resno 3
:resname 1_supplekey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 325
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 325
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 325
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 325
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 325
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 325
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 325
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 325
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 325
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 325
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 325
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

        :varattno 15
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 15
        :location 325
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 325
    }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDecls <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :location 289
    }
    }
    :location 224
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 1
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggsplit 0
    :aggno -1
    :aggtransno -1
    :location 137
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
    )
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause (

```

```

{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: pull_up_sublinks - File: src/backend/optimizer/prop/prepjointree.c, Relation: N/A, Explanation: Initiating pull-up of sublinks (ANY/EXISTS)

WARNING: ===== after pull up and before planning path list =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
}
)

```

```

        :location -1
    }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: ===== Sorted Qual List in order_qual_clauses =====
DETAIL: (
  (RESTRICTINFO
    :clause
    {OEXPR
      :opno 674
      :opfuncid 297
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        (VAR
          :varno 1
          :varattno 6
          :vartype 701
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        )
        (CONST
          :consttype 701
          :consttypmod -1
          :constcollid 0
          :constlen 8
          :constbyval true
          :constnull false
          :location -1
          :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
        )
      )
    }
    :location -1
  )
  :is_pushed_down true
  :can_join false
  :pseudoconstant false
  :has_clone false
  :is_clone false
  :leakproof false
  :has_volatile 0
  :security_level 0
  :num_base_rels 1
  :clause_relids (b 1)
  :required_relids (b 1)
  :incompatible_relids (b)
  :outer_relids (b)
  :left_relids (b 1)
  :right_relids (b)
  :orclause <>
  :rinfo_serial 2
  :eval_cost.startup 0
  :eval_cost.per_tuple 0.0025
  :norm_selec 0.45285077010838557
  :outer_selec -1
  :mergeopfamilies <>
  :left_em <>
  :right_em <>
  :outer_is_left false
  :hashjoinoperator 0
  :left_bucketsize -1
  :right_bucketsize -1
  :left_mcvfreq -1
  :right_mcvfreq -1
  :left_hasheqoperator 0
  :right_hasheqoperator 0
}
  (RESTRICTINFO
    :clause
    {OEXPR
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        (VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location -1
        )
        (PARAM
          :paramkind 1
          :paramid 0
          :paramtype 23
          :paramtypmod -1
          :paramcollid 0
          :location -1
        )
      )
    }
    :location -1
  )
  :is_pushed_down true
  :can_join false
  :pseudoconstant false
  :has_clone false
  :is_clone false
  :leakproof false
  :has_volatile 2
  :security_level 0
  :num_base_rels 1
  :clause_relids (b 1)
  :required_relids (b 1)
  :incompatible_relids (b)

```

```

:outer_relds (b)
:left_relds (b 1)
:right_relds (b)
:orclause <>
:rinfo_serial 1
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-06
:outer_selec -1
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:em_relds (b 1)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1)
}
}
:right_em
{EQUIVALENCEMEMBER
:em_expr
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
:em_relds (b)
:em_is_const true
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relds (b 1)
}
}
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}

```

WARNING: ===== create_seqscan_plan scan->plan.qual =====

```

DETAIL: {
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
}

```

WARNING: ===== after pull up and before planning path list =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
}

```

```

:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias
      {ALIAS
        :aliasname c
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname c
        :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
          c_acctbal" "c_mktsegment" "c_comment")
      }
    :rtekind 0
    :relid 16546
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals (
      {OPEXPR
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 6
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 6
            :location -1
          }
          {CONST
            :consttype 701
            :consttypmod -1
            :constcollid 0
            :constlen 0
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 8 [ 0 0 0 0 -120 -77 64 ]
          }
        )
        :location -1
      }
    )
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: ===== create_seqscan_plan scan->plan.qual =====

```

DETAIL: (
  {OPEXPR
    :opno 674
    :opfuncid 297
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR

```

```

:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
}
:location -1
}
)

```

WARNING: ===== securityQuals after SS_process_sublinks =====

```

DETAIL: ( (
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 5922
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
)
}
))

```

WARNING: ===== after pull up and before planning path list =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
}
)
}

```

```

:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
  {ALTERNATIVESUBPLAN
    :subplans (
      {SUBPLAN
        :subLinkType 0
        :testexpr <>
        :paramids <>
        :plan_id 1
        :plan_name SubPlan\ 1
        :firstColType 2278
        :firstColTypmod -1
        :firstColCollation 0
        :useHashTable false
        :unknownEqFalse true
        :parallel_safe false
        :setParam <>
        :parParam (i 0)
        :args (
          {VAR
            :varno 1
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location -1
          }
        )
        :startup_cost 0
        :per_call_cost 5922
      }
      {SUBPLAN
        :subLinkType 2
        :testexpr
        {OPEXPR
          :opno 96
          :opfuncid 65
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 1
              :varattno 2
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 1
              :varattnosyn 2
              :location -1
            }
            {PARAM
              :paramkind 1
              :paramid 1
              :paramtype 23
              :paramtypmod -1
              :paramcollid 0
              :location -1
            }
          )
        }
        :location -1
      }
    )
    :paramids (i 1)
    :plan_id 2
    :plan_name SubPlan\ 2
    :firstColType 23
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable true
    :unknownEqFalse true
    :parallel_safe true
    :setParam <>
    :parParam <>
    :args <>
    :startup_cost 5716.52
    :per_call_cost 0.0025
  }
)
)
)
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 116
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true

```

```

:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 187
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 207
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 199
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 228
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 240
}
)
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 387
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 400
}
)
:location 398
}
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 12
    :location 427
  }
  {VAR
    :varno 3
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 13
    :location 442
  }
)
:location 440
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 6
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes <>
      :aggdirectargs <>
      :args <>
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggssplit 0
      :aggno -1
      :aggtransno -1
      :location 137
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

WARNING: ===== Relation orders baserestrictinfo security levels =====
WARNING: Qual security_level: 0
WARNING: Qual clause:
DETAIL: (ALTERNATIVESUBPLAN
:subplans (
  {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramIds <>
  }
)

```

```

:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (1 0)
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
)
:startup_cost 0
:per_call_cost 5922
}
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
}
)
}

```

WARNING: ===== Sorted Qual List in order_qual_clauses =====:

```

DETAIL: {
{RESTRICINFO
:clause
{OEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 187
  }
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 207
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
  }
)
:location 199
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relids (b 1)
:required_relids (b 1)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 1)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6922656734118627
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1

```

```

:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
{RESTRICTINFO
:clause
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstCollType 2278
:firstCollTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 5922
}
{SUBPLAN
:sublinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstCollType 23
:firstCollTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
)
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relds (b 1)
:required_relds (b 1)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b)
:right_relds (b)
:orclause <>
:rinfo_serial 1
:eval_cost.startup 0
:eval_cost.per_tuple 5922
:norm_selec 0.5
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
{RESTRICTINFO
:clause
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 228
}
)
}

```

```

(CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
)
:location 240
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relsids (b 1)
:required_relsids (b 1)
:incompatible_relsids (b)
:outer_relsids (b)
:left_relsids (b 1)
:right_relsids (b)
:orclause <>
:rinfo_serial 3
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.5
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
)

WARNING: ===== create_seqscan_plan scan->plan.qual =====
DETAIL: (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 187
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 207
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 199
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEq false true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 5922
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0

```

```

        :location -1
      }
    }
    :location -1
  }
  :paramIds (i 1)
  :plan_id 2
  :plan_name SubPlan\ 2
  :firstColType 23
  :firstColTypeMod -1
  :firstColCollation 0
  :useHashTable true
  :unknownEqFalse true
  :parallel_safe true
  :setParam <>
  :parParam <>
  :args <>
  :startup_cost 5716.52
  :per_call_cost 0.0025
}
}
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 228
  }
  {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
  }
)
:location 240
}
}
)

WARNING: ===== create_seqscan_plan scan->plan.qual =====
DETAIL: (
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 12
    :location 427
  }
  {VAR
    :varno 3
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 13
    :location 442
  }
)
:location 440
}
)

WARNING: === PLANNER SUGGESTION ===
Plan:
GroupAggregate (cost=8883274415.71..8883274719.76 rows=5 width=17)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  -> Sort (cost=8883274415.71..8883274517.04 rows=40534 width=9)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    -> Hash Semi Join (cost=212549.34..8883271313.47 rows=40534 width=9)
      Output: orders.o_orderpriority
      Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
      -> Seq Scan on public.orders (cost=0.00..8883048508.00 rows=144199 width=13)
        Output: orders.o_orderpriority, orders.o_orderkey
        Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).coll1)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zon
e))
        SubPlan 2
        -> Seq Scan on public.customer c (cost=0.00..5547.00 rows=67808 width=4)
          Output: c.c_custkey
          Filter: (c.c_acctbal > '5000'::double precision)
        -> Hash (cost=179731.50..179731.50 rows=2000307 width=4)
          Output: lineitem.l_orderkey
          -> Seq Scan on public.lineitem (cost=0.00..179731.50 rows=2000307 width=4)
            Output: lineitem.l_orderkey
            Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
=====
WARNING: === EXECUTOR EXECUTION ===
Rows Processed: 5
Plan & Stats:
GroupAggregate (cost=8883274415.71..8883274719.76 rows=5 width=17) (actual time=2643.564..2645.319 rows=5.00 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=3 read=134400, temp read=14748 written=14748
  -> Sort (cost=8883274415.71..8883274517.04 rows=40534 width=9) (actual time=2643.113..2643.929 rows=23317.00 loops=1)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    Sort Method: quicksort Memory: 769kB
    Buffers: shared hit=3 read=134400, temp read=14748 written=14748
    -> Hash Semi Join (cost=212549.34..8883271313.47 rows=40534 width=9) (actual time=1837.558..2636.273 rows=23317.00 loops=1)
      Output: orders.o_orderpriority
      Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
      Buffers: shared read=134400, temp read=14748 written=14748
      -> Seq Scan on public.orders (cost=0.00..8883048508.00 rows=144199 width=13) (actual time=48.291..404.281 rows=25383.00 loops=1)
        Output: orders.o_orderpriority, orders.o_orderkey
        Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).coll1)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zon
e))
        Rows Removed by Filter: 1474617
        Buffers: shared read=29680
        SubPlan 2
        -> Seq Scan on public.customer c (cost=0.00..5547.00 rows=67808 width=4) (actual time=0.192..33.895 rows=67989.00 loops=1)

```

```
Output: c.c_custkey
Filter: (c.c_acctbal > '5000')::double precision)
Rows Removed by Filter: 82011
Buffers: shared read=3672
-> Hash (cost=179731.50..179731.50 rows=2000307 width=4) (actual time=1783.430..1783.430 rows=3793296.00 loops=1)
Output: lineitem.l_orderkey
Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6223kB
Buffers: shared read=104720, temp written=10750
-> Seq Scan on public.lineitem (cost=0.00..179731.50 rows=2000307 width=4) (actual time=0.559..1170.362 rows=3793296.00 loops=1)
Output: lineitem.l_orderkey
Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
Rows Removed by Filter: 2207919
Buffers: shared read=104720
=====
```

QUERY PLAN

```
[
+
{
+
"Plan": {
+
"Node Type": "Aggregate",
+
"Strategy": "Sorted",
+
"Partial Mode": "Simple",
+
"Parallel Aware": false,
+
"Async Capable": false,
+
"Startup Cost": 8883274415.71,
+
"Total Cost": 8883274719.76,
+
"Plan Rows": 5,
+
"Plan Width": 17,
+
"Actual Rows": 5.00,
+
"Actual Loops": 1,
+
"Disabled": false,
+
"Output": ["orders.o_orderpriority", "count(*)"],
+
"Group Key": ["orders.o_orderpriority"],
+
"Plans": [
+
{
+
"Node Type": "Sort",
+
"Parent Relationship": "Outer",
+
"Parallel Aware": false,
+
"Async Capable": false,
+
"Startup Cost": 8883274415.71,
+
"Total Cost": 8883274517.04,
+
"Plan Rows": 40534,
+
"Plan Width": 9,
+
"Actual Rows": 23317.00,
+
"Actual Loops": 1,
+
"Disabled": false,
+
"Output": ["orders.o_orderpriority"],
+
"Sort Key": ["orders.o_orderpriority"],
+
"Sort Method": "quicksort",
+
"Sort Space Used": 769,
+
"Sort Space Type": "Memory",
+
"Plans": [
+
{
+
"Node Type": "Hash Join",
+
"Parent Relationship": "Outer",
+
"Parallel Aware": false,
+
"Async Capable": false,
+
"Join Type": "Semi",
+
"Startup Cost": 212549.34,
+
"Total Cost": 8883271313.47,
+
"Plan Rows": 40534,
+
"Plan Width": 9,
+
"Actual Rows": 23317.00,
+
"Actual Loops": 1,
+
"Disabled": false,
+
"Output": ["orders.o_orderpriority"],
+
"Inner Unique": false,
+
"Hash Cond": "(orders.o_orderkey = lineitem.l_orderkey)",
+
"Plans": [
+
{
+
"Node Type": "Seq Scan",
+
"Parent Relationship": "Outer",
+
"Parallel Aware": false,
+
"Async Capable": false,
+
"Relation Name": "orders",
+
"Schema": "public",
+
"Alias": "orders",
+
"Startup Cost": 0.00,
+
"Total Cost": 8883048508.00,
+
"Plan Rows": 144199,
+
"Plan Width": 13,
+

```

```

+         "Actual Rows": 25383.00,
+
+         "Actual Loops": 1,
+
+         "Disabled": false,
+
+         "Output": [{"orders.o_orderpriority", "orders.o_orderkey"}],
+
+         "Filter": "((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).col1)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))",
+
+         "Rows Removed by Filter": 1474617,
+
+         "Plans": [
+
+             {
+
+                 "Node Type": "Seq Scan",
+
+                 "Parent Relationship": "SubPlan",
+
+                 "Subplan Name": "SubPlan 2",
+
+                 "Parallel Aware": false,
+
+                 "Async Capable": false,
+
+                 "Relation Name": "customer",
+
+                 "Schema": "public",
+
+                 "Alias": "c",
+
+                 "Startup Cost": 0.00,
+
+                 "Total Cost": 5547.00,
+
+                 "Plan Rows": 67808,
+
+                 "Plan Width": 4,
+
+                 "Actual Rows": 67989.00,
+
+                 "Actual Loops": 1,
+
+                 "Disabled": false,
+
+                 "Output": [{"c.c_custkey"}],
+
+                 "Filter": "(c.c_acctbal > '5000'::double precision)",
+
+                 "Rows Removed by Filter": 82011
+
+             }
+
+         ],
+
+     {
+
+         "Node Type": "Hash",
+
+         "Parent Relationship": "Inner",
+
+         "Parallel Aware": false,
+
+         "Async Capable": false,
+
+         "Startup Cost": 179731.50,
+
+         "Total Cost": 179731.50,
+
+         "Plan Rows": 2000307,
+
+         "Plan Width": 4,
+
+         "Actual Rows": 3793296.00,
+
+         "Actual Loops": 1,
+
+         "Disabled": false,
+
+         "Output": [{"lineitem.l_orderkey"}],
+
+         "Hash Buckets": 262144,
+
+         "Original Hash Buckets": 262144,
+
+         "Hash Batches": 32,
+
+         "Original Hash Batches": 16,
+
+         "Peak Memory Usage": 6223,
+
+         "Plans": [
+
+             {
+
+                 "Node Type": "Seq Scan",
+
+                 "Parent Relationship": "Outer",
+
+                 "Parallel Aware": false,
+
+                 "Async Capable": false,
+
+                 "Relation Name": "lineitem",
+
+                 "Schema": "public",
+
+                 "Alias": "lineitem",
+
+                 "Startup Cost": 0.00,
+
+                 "Total Cost": 179731.50,
+
+                 "Plan Rows": 2000307,
+
+                 "Plan Width": 4,
+
+                 "Actual Rows": 3793296.00,
+
+                 "Actual Loops": 1,
+
+                 "Disabled": false,
+
+                 "Output": [{"lineitem.l_orderkey"}],
+
+                 "Filter": "(lineitem.l_commitdate < lineitem.l_receiptdate)",
+
+                 "Rows Removed by Filter": 2207919
+
+             }
+
+         ]
+
+     }
+
+ ]
+
+ }
+
+ ]
+
+ }
+
+ ]
+
+ ]

```